Solvers

Grid and spectral

JAX-CFD	PhiFlow
XLB	PICT
Warp-NS	Exponax
INS.jl	OpenFOAM

Finite element

FEniCS	Firedrake
JAX-FEM	torch-fem
TopOpt.jl	deal.II

Interface

Tesseract

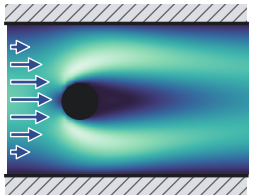
`apply(x) → y`

`vjp(x, v) → g`

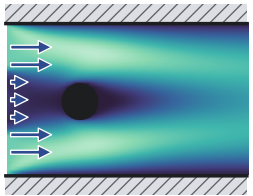
one container per solver

Tasks

Forward solve



Optimized inflow



plus H, S and F3 tasks

Results

2D Navier–Stokes, checks per suite

	Fwd	Cost	Grad	Opt.
JAX-CFD	–	✓	✓	–
PhiFlow	✓	✓	✓	–
INS.jl	✓	✓	✓	✓
XLB	✓	✓	✓	✓
PICT	–	–	✓	✓
Warp-NS	–	✓	–	–
Exponax	–	✓	✓	✓
OpenFOAM	–	✓		

✓ pass – partly not applicable